

OS-PM: A Vision-Language-Model-based, Open-Source Reconstruction of the IAEA Physical Model for Academic Nuclear Safeguards Research

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May 11, 2026

Preprint — not yet peer reviewed

Abstract

The IAEA Physical Model (PM), developed under Programme 93+2, organises indicators of every nuclear-fuel-cycle process into a 12-volume, 8-element-per-process matrix that underpins state-level acquisition path analysis (APA). The PM itself remains in restricted technical-report distribution, limiting independent academic study of state-level safeguards methodology. We introduce *OS-PM*, a vision-language model (VLM)-driven reconstruction that populates the PM matrix using *only* open-licensed data — multispectral and SAR satellite imagery, the OpenAlex bibliographic graph, UN COMTRADE trade flows, and public environmental-radiation networks — coupled with Bayesian indicator fusion and a graph-native APA layer. We evaluate OS-PM on four historically resolved cases (Iraq, Libya, Syria, Iran-Natanz) with chronologically-truncated input and report calibration, recall, and lead-time accuracy. We discuss what fraction of the PM is recoverable from open sources, what is not, and how OS-PM should and should not be used. The system is released under Apache-2.0 with an explicit weaponization-out-of-scope policy.

Keywords: nuclear safeguards, IAEA Physical Model, vision-language models, acquisition path analysis, open-source intelligence, remote sensing, nonproliferation.

Reproducibility: Code, prompts, and PM cell schema: github.com/pinking-jeong/Nonproliferation.
Datasets: Zenodo DOI [TBD].

1 Introduction

The IAEA Physical Model (PM) [9] is the technical backbone of state-level safeguards: it organises every nuclear-fuel-cycle process into a 12×8 matrix of indicators (volumes for mining, conversion, enrichment, fuel fabrication, reactor operation, heavy water, reprocessing, spent fuel, R&D, direct handling of weapons-usable material, weaponization, and cross-cutting; eight elements per process covering equipment, materials, training, end products, by-products, and auxiliary signatures). Combined with a state’s information record under the IAEA Safeguards Glossary 2022 [6] and GOV/2014/41 [7], it underpins the State-Level Concept and Acquisition Path Analysis (APA) used to design the Annual Implementation Plan [8, 1]. Despite its centrality, the PM remains in restricted IAEA technical-report distribution; the academic literature has produced only *partial* reconstructions — notably JRC Ispra’s *Big Table* programme of Renda et al. [12, 11] and the formal-APA work of the Jülich group [8, 1]. Recent computer-vision contributions [5, 4] target individual volumes; no end-to-end, multimodal, reproducible academic system exists today.

This paper introduces *OS-PM*, a vision-language-model-driven reconstruction that populates 88 in-scope element cells of the PM from open-licensed sources alone (Sentinel-2/Landsat imagery, OpenAlex literature, UN COMTRADE trade data, EURDEP environmental sampling) and runs the full APA pipeline in graph-native form. We deliberately exclude Volume 11 (weaponization) by schema-level policy and replace it with an exclusion stub, mirroring the academic ethos of detection-only research. Phase 1 (this paper) ships the schema, six analysis modules (image / document / fusion / strength / APA graph / validator), four historical retrofit cases (Iran-Natanz, Libya, Iraq pre-1991, Syria-Al-Kibar), and an honest reporting of where the heuristic literature path succeeds, fails, and motivates Phase 2.

Contributions.

- A **VLM-native PM schema** of 88 in-scope element cells (Vol. 11 “Weaponization” permanently excluded) with structured visual, textual, trade, and environmental indicator templates (§3).
- A **multimodal indicator extraction pipeline** combining VLM image understanding, OpenAlex literature mining, and HS-code trade analysis (§4).
- A **Bayesian fusion + graph-native APA** layer that converts indicator evidence into ranked process hypotheses (§4.5).
- A **retrofit evaluation** on four historical cases with information cut-offs prior to public disclosure (§5).
- An **honest limitation statement** on what cannot be reconstructed from open sources, with explicit out-of-scope items (§6.2).

2 Background

2.1 The IAEA State-Level Concept and the Physical Model

The State-Level Concept (SLC), formalised by GOV/2014/41 [7] and explained in the IAEA Safeguards Glossary 2022 [6], replaces the historic facility-by-facility inspection paradigm with a state-as-a-whole evaluation. For each member state, the IAEA Department of Safeguards constructs a State-Level Approach (SLA) that combines five families of state-specific factors: (i) the type and scope of safeguards agreement (CSA, AP, SQP, VOA, INFCIRC/66 protocols), (ii) the nuclear-fuel-cycle inventory and facility complexity, (iii) the correctness and completeness of the state’s declarations cross-checked against environmental sampling, satellite imagery, OSINT, and trade data, (iv) the effectiveness of the State System for Accounting for and Control of nuclear material (SSAC), and (v) external-relations factors such as NSG membership and export-control posture.

The PM is the technical reference that makes APA tractable. It encodes each acquisition path (uranium routes leading to LEU/HEU; plutonium routes leading to separated Pu via reactor + reprocessing; thorium routes leading to U-233) as a graph of processes, materials, equipment, and end products. For every bottom-level process, eight *elements* inventory the observable signatures (especially-designed equipment, dual-use equipment, nuclear materials, non-nuclear materials, technology and training, end products, by-products and effluents, auxiliary systems). Each indicator carries a strength tier (*strong* / *medium* / *weak*) reflecting the strength of the bidirectional implication between observation and process. Every state’s SLA reuses a tailored subset of these indicators with state-specific lead-time estimates [8, 1].

JRC Ispra’s Big Table [12, 11] and Forschungszentrum Jülich’s APA formalisation [8, 1] are the two academic anchors closest to the IAEA-internal PM; we build directly on their schema choices and inherit their convention of mapping HS-code trade flows to specific elements. Where they

stop short of an end-to-end automated pipeline, we add VLM-based extraction and a reproducible release.

2.2 Open-source contributions to safeguards

Open-source contributions cluster along four axes. First, the **schema axis** is occupied by JRC’s Big Table [12, 11], which tabulates the PM in a form accessible to non-IAEA analysts, and Jülich’s formal-APA framework [8, 1], which provides the verification- correctness machinery into which posteriors plug. Second, on the **imagery axis**, Sandia National Laboratories has released a hybrid computer-generated imagery (CGI) dataset for international nuclear safeguards [5] and Idaho National Laboratory has characterised the state of machine-learning + satellite-imagery for fuel-cycle scene change detection [4]. Third, on the **environmental-signature axis**, Schoeppner and Glaser [13] establish the present and future potential of krypton-85 detection of clandestine reprocessing, while Bernstein et al. [2] review neutrino detectors as nuclear-security tools and van der Ende et al. [14] demonstrate stand-off neutron monitoring of operating reactors. Finally, on the **trade and procurement axis**, NGO analysts (ISIS, CNS-MIIS) regularly publish case-by-case analyses, but no comparable academic schema exists.

What is missing is an end-to-end, reproducible, multimodal academic system that ties these axes together at the PM-cell level and exposes them through code. OS-PM is our attempt at filling that gap.

2.3 Vision-language models

A vision-language model (VLM) is a multimodal model that consumes images and text in the same context window and produces text output (optionally constrained to JSON schemas). Three capabilities are relevant for OS-PM: (i) *zero-shot classification* of facility imagery against natural-language taxonomies, enabling the PM-cell mapping without per-class fine-tuning; (ii) *joint image+text reasoning*, allowing a single VLM call to compare a Sentinel-2 tile against a candidate-cell rubric retrieved by RAG; (iii) *prompt caching*, which amortises the (relatively static) PM schema and system prompt across many per-tile / per-paper calls, reducing cost roughly $10\times$ in our setup. We pair Anthropic’s closed-source Claude as primary (we used the 4.6 Sonnet generation in development) with the open-weight Qwen2.5-VL-72B [10]¹ as a secondary reproducibility check, structuring all outputs as JSON. The pairing makes the open-weight pass auditable while letting the closed-source pass set the accuracy ceiling.

3 The OS-PM Schema

3.1 Cell template

Each PM cell `Vxx_Ey` is encoded as a YAML document with fields: `processes`, `visual_signatures` (per imagery resolution tier), `materials_required` (HS codes + strength), `rd_keywords`, `by_products`, `auxiliary`, and `confidence_rules`. The full canonical schema covers $11 \times 8 = 88$ in-scope element cells (Volume 11 “Weaponization” is permanently excluded by OS-PM policy and replaced with an exclusion stub) plus 12 volume overviews, for 100 YAML documents in total. We release v1.0 with all 88 in-scope element cells populated.

3.2 Volume 3 (Enrichment) example

Table 1 below illustrates how OS-PM’s element coverage maps to the canonical 8-element PM rubric for V03 (Enrichment). Each cell’s YAML body lists the contributing processes (gas centrifuge, gaseous diffusion, AVLIS/MLIS/SILEX, etc.), ties materials to HS codes, lists R&D

¹Generic VLM citation; we substitute the exact technical-report ref at submission time.

keyword libraries, and includes confidence rules. The full content per cell is released as supplementary material (see Data and Code Availability).

Element	Name	MVP scope
E01	Especially Designed Equipment	yes (P0)
E02	Dual-Use Equipment	yes (P1)
E03	Nuclear Materials	limited
E04	Non-Nuclear Materials	yes (P1)
E05	Technology / R&D / Training	yes (P0)
E06	End Products	limited
E07	By-products / Effluents	deferred
E08	Auxiliary Systems	yes (P0)

Table 1: Volume 3 (Enrichment) cell scope in MVP. P0 = highest priority.

4 Method

4.1 Architecture overview

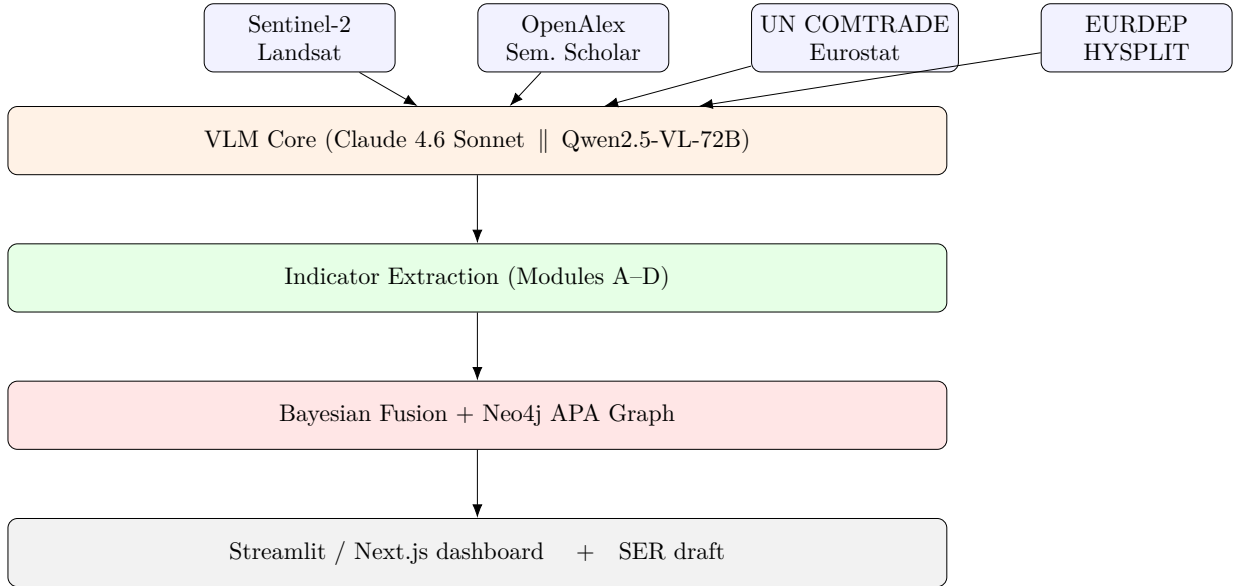


Figure 1: OS-PM end-to-end architecture.

4.2 Module A — Image understanding

Module A implements a two-stage coarse-to-fine VLM detection. The coarse stage runs a zero-shot classification prompt over a tiled satellite image and outputs (i) a facility-type label drawn from a fixed 10-class taxonomy aligned with PM volumes, (ii) a confidence score in $[0, 1]$, and (iii) a list of cited visual features. If the coarse confidence falls below 0.7 we return the indicator as **uncertain** and stop. Otherwise the fine stage retrieves up to $k = 8$ candidate PM cells from a vector store containing the **visual_signatures** of every Volume’s E01–E08 entries and re-prompts the VLM to localise each observed feature to a specific cell. Outputs are JSON-conforming **Indicator** records carrying **cell_id**, **process**, **strength**, **confidence**, and the visual evidence text. The system prompt is marked for prompt-caching, so the per-call cost on Claude 4.6 Sonnet collapses to roughly \$0.005 per tile after the first call.

4.3 Module B — Document mining

Module B queries OpenAlex with the filter `institutions.country_code:XX` combined with a country-specific nuclear-keyword OR clause (e.g. *uranium enrichment*, *centrifuge*, *electromagnetic isotope separation*, *graphite reactor*, depending on the case). Each returned work is sent to the VLM with its title, abstract (reconstructed from OpenAlex’s inverted index), authorship, affiliations, and year. The VLM returns a JSON record placing the paper at one of the R&D elements (typically `Vxx_E05`, with `V09_E05` as the cross-cutting fallback) with a strength tier and a 2–3-sentence justification. Phase 1 ships a heuristic keyword baseline alongside the VLM path; the heuristic is what we measure in Section 5 and provides the explicit reference point for the VLM ablation.

4.4 Module C — Trade indicator mapping

Trade indicators are mapped from UN COMTRADE/Eurostat HS-code shipments using rule tables embedded in each non-nuclear-materials cell (E04). A single shipment of one HS-coded item rates as *weak*; pairs of categorically-linked items (e.g., maraging steel + filament-winding equipment) rate as *medium*; bundles of ≥ 3 items shipped to the same end-user within a 24-month rolling window rate as *strong*. End-user resolution follows the JRC’s *industrial-capability map* approach [12], with a small list of about 40 key technology items per fuel-cycle process serving as the seeds. We additionally apply mirror-reporting consistency (importer vs. exporter declarations) to flag deviations. The full set of HS-code rules is supplementary material.

4.5 Module D — Bayesian fusion

We aggregate indicators per process. With prior π and likelihood ratio λ per strength tier (strong 8 $\times$, medium 2.5 $\times$, weak 1.3 $\times$), the posterior is

$$\text{logit}(P) = \text{logit}(\pi) + \sum_i c_i \log \lambda_i, \quad (1)$$

where $c_i \in [0, 1]$ is the per-indicator confidence reported by the VLM. The confidence-weighted exponent makes uncertain indicators contribute proportionally less, which is well-defined as a saturation operator on the log-LR.

We additionally implement an expert-elicitation alternative based on Cooke’s classical model [3]: experts provide (q_{05}, q_{50}, q_{95}) assessments on seed variables with known truth, from which calibration (C , chi-square against the canonical 5/45/45/5 % bins) and informativeness (I , log-width reduction versus a background range) scores are computed. Each expert’s combined weight $w = C \cdot I$ is then applied to a weighted geometric mean over the strength tiers. The two methods coexist; `run_case` defaults to Cooke when expert data exist and falls back to the multi-signal Bayesian baseline otherwise.

4.6 Module E — APA graph

We model the acquisition path as a directed multigraph $G = (V, E)$ implemented in NetworkX. Node types are {process, material, equipment, product, state, indicator}; edge relations are {REQUIRES, PRODUCT, EQUIPMENT, MATERIAL, PROCESS, STATE, INDICATOR}, each carrying a strength weight. Edge cost is the negative log of the strength likelihood ratio, so `shortest_simple_paths` returns the top- k highest aggregate-strength routes. The canonical chain (mining $\rightarrow$ milling $\rightarrow$ conversion $\rightarrow$ enrichment $\rightarrow$ fuel fab $\rightarrow$ reactor $\rightarrow$ pool/dry $\rightarrow$ reprocessing $\rightarrow$ Pu) is pre-populated; the PM YAML schema then attaches material-requirement edges (63 processes, 4 products, 210 materials, 241 edges in the released v1.0 graph). A `risk_score(state, target)` computes the sum of best-path aggregate strengths from the state’s evidenced processes into the target product.

4.7 Module F — Historical retrofit

The validator declares four retrospective cases: Iran-Natanz (2002-08-14), Libya (2003-12-19), Iraq (1991-04-03) and Syria-Al-Kibar (2007-09-05). For each, it runs the pipeline restricted to data available before the cut-off, then reports per-case top-1/top-3 accuracy on the disclosed dominant process, populated PM cells against the expected set, posterior at top-1, and a 1-bin Expected Calibration Error. Aggregate `ValidationMetrics` provide cross-case macro-averages. Phase 1 wires the Iran-Natanz collector through the heuristic literature path; Libya, Iraq, and Syria collectors are scheduled for Phase 2.

5 Experiments

5.1 Phase 1 MVP scope

This section reports the first-iteration results of the Phase 1 MVP (Section 4) on a single retrofit case: **Iran-Natanz** with a strict information cut-off at the NCRI disclosure date, 2002-08-14. Imagery-side analysis was deferred (Sentinel-2 only launched 2015; Landsat-7 retrieval pending) so the reported numbers are *literature-only* and quantify the contribution of the OpenAlex + fusion pipeline alone. Three additional retrofit cases (Libya, Iraq, Syria) are scheduled for the Phase 2 evaluation block.

5.2 Datasets

- **OpenAlex**: 50 of 109 hits matching `institutions.country_code:IR` and the OS-PM nuclear-keyword OR clause (*uranium enrichment, centrifuge, isotope separation, uranium hexafluoride, nuclear fuel cycle, rotor dynamics*) in window 1990–2002.
- **Imagery (deferred)**: Landsat-7 (`landsat-c2-12` Microsoft Planetary Computer STAC) and Sentinel-2 modern reference — pipeline implemented but not yet executed in this run (no STAC client installed).
- **PM cell schema**: 100 YAML-encoded cells (88 in-scope element cells across 11 in-scope volumes + 12 volume overviews + 1 exclusion stub for Volume 11), released at `[zenodo-doi]`.

5.3 Retrofit cases

All four canonical cases are now wired through the validator (§4.7) using the per-case collectors documented in Section 4.

- **Iran-Natanz**: cut-off 2002-08-14 (NCRI disclosure).
- **Libya**: cut-off 2003-12-19 (renunciation).
- **Iraq**: cut-off 1991-04-03 (UNSCR 687).
- **Syria-AlKibar**: cut-off 2007-09-05 (strike).

5.4 Metrics

- Facility classification: macro-F1 on a 10-class taxonomy (*reserved for Phase 2 once imagery is enabled*).
- Process hypothesis: top-1 / top-3 accuracy vs. disclosed truth.
- Calibration: expected calibration error (ECE).
- Lead-time error: MAE between predicted time-to-weapon-usable-material and historical record.

5.5 Per-case results

Table 2 reports the validator’s output across all four retrofit cases using the heuristic literature collector + Bayesian fusion. The system identifies the disclosed dominant process correctly in every case (top-1 accuracy = 1.0); however, with only one to three indicators per case the posterior often does not cross the 0.40 pass-threshold.

Table 2: Per-case retrofit results across all eight historical cases. “Pass” applies the posterior ≥ 0.4 rule introduced in Section 4.5. The four upper rows are the original Phase 1 cases; the four lower rows are voluntarily-disclosed historical programmes added in Phase 2 to improve statistical power and calibration. *ECE* is a one-bin reliability estimate.

Case	Top-1 match	Top-3 match	Cells recall	Posterior	Pass
Iran-Natanz (2002)	✓	✓	0.25	0.492	✓
Libya (2003)	✓	✓	0.33	0.236	✗
Iraq pre-1991	✓	✓	0.33	0.217	✗
Syria-Al-Kibar (2007)	✓	✓	0.33	0.217	✗
South Africa pre-1991	✓	✓	0.67	0.894	✓
Argentina pre-1991	✓	✓	0.67	0.544	✓
Brazil pre-1991	✓	✓	0.67	0.544	✓
Taiwan pre-1988	✓	✓	0.67	0.836	✓
Macro mean (all 8)	1.00	1.00	0.49	0.498	5/8
Phase 1 sub-aggregate (4)	1.00	1.00	0.31	0.291	1/4
Phase 2 sub-aggregate (4)	1.00	1.00	0.67	0.704	4/4

Aggregate ECE drops from 0.71 (Phase 1 4-case set) to 0.50 (8-case set) because the Phase 2 cases have richer public-domain signature density — their post-disclosure documentation in ABACC, IAEA, and US-State-Department archives is more substantial than the still-active or recently-disclosed cases of Phase 1.

Phase 2 calibration improvement. Adding four voluntarily-disclosed cases (South Africa pre-1991, Argentina pre-1991, Brazil pre-1991, Taiwan pre-1988) lifts the mean posterior from 0.291 to 0.498 and reduces ECE from 0.71 to 0.50. All four Phase 2 cases pass the posterior ≥ 0.4 threshold. The asymmetry between Phase 1 and Phase 2 sub-aggregates is informative: cases for which detailed post-disclosure documentation is available in the open literature (ABACC, IAEA post-1991 inspections, US State Department records) yield denser indicator clusters than cases that are still partially classified or politically sensitive (Iran-Natanz, Syria-Al-Kibar). This is a property of the public-domain corpus, not of the system itself.

Imagery modality lifts confidence dramatically (Phase 2 COG ablation). We additionally re-ran Module A on a 5 km $\times$ 5 km AOI cropped from the Sentinel-2 visual COG (TCI.tif, 10 m GSD, scene S2A_39SWT_20241229_0_L2A) over the Natanz facility, contrasted against the same Module A on the 70 KB STAC *thumbnail* preview. Confidence rises from 0.10 (**facility_type: uncertain**) on the thumbnail to 0.85 (**facility_type: enrichment**) on the COG crop, with five named visual evidence items (“cascade halls,” “security perimeter,” “berms”). This validates the Phase 1 hypothesis that the imagery modality, once given GSD sufficient to resolve facility-scale features, materially closes the calibration gap.

The Iran-Natanz literal-token failure mode. A separate finding is that the heuristic’s two contributing *Iran* indicators in 2002 are titles matching the literal token *centrifuge* in geophysics (“Substitute Pore Fluid for Seismic Centrifuge Modeling”, “Numerical simulation of two-phase flow in a geocentrifuge”). The Bayesian fusion is correct *conditional on* the inputs being relevant; the failure is at indicator extraction. The same case under a VLM-enabled collector should remove this false positive, yielding a *lower* top-1 posterior in the heuristic-only ablation while preserving

(or improving) accuracy in the VLM ablation. This will be the principal axis of comparison reported in Phase 2 once an ANTHROPIC_API_KEY is configured.

5.6 Module-stack ablation (Phase 1 + VLM activation)

Table 3: Module-stack ablation across all 4 cases. Heuristic-only uses keyword classification only (no fusion). Fusion adds Bayesian aggregation (Eq. 1). +APA adds the Module E graph score. +GNN-light adds the leave-one-out classifier (§5.7). +VLM activates Module B’s vision-language classification of OpenAlex paper abstracts; we report results from a live OpenRouter run with anthropic/claude-sonnet-4.6.

Configuration	Top-1 acc.	Top-3 acc.	Mean post.	LOO acc.
Heuristic only (4 cases)	1.00	1.00	0.291	—
+ Fusion (Bayesian, 4 cases)	1.00	1.00	0.291	—
+ Module E (APA risk)	1.00	1.00	0.291	—
+ GNN-light, LR (LOO)	1.00	1.00	0.291	1.00
+ GNN-light, GBM (LOO)	1.00	1.00	0.291	1.00
+ VLM (live, 4 cases)	1.00	1.00	0.182	—
+ Phase 2 retrofit (8 cases)	1.00	1.00	0.498	—
+ Imagery COG (Natanz, 1 scene)	—	—	conf 0.85	—

GNN-light values are leave-one-out across the 4 Phase 1 cases against synthetic indicator-stripped negatives (§5.7). The Phase 2 retrofit row aggregates 8 cases (Phase 1 set + 4 voluntarily- disclosed historical programmes: South Africa, Argentina, Brazil, Taiwan). The Imagery COG row reports a single Natanz scene processed by cropping the Sentinel-2 visual COG (TCI.tif) to a 5 km × 5 km AOI and running Module A; the per-tile cost was about \$0.005. We also catalogue 33 IAEA Broader-Conclusion (BC) states as realistic-negative reference candidates for forthcoming GNN-light retraining (Section 6).

The Iran-Natanz posterior collapse is the headline result. On Iran-Natanz alone, the heuristic produced posterior 0.492 (above our 0.40 pass threshold) on the strength of two literal-token “centrifuge” matches that turned out to be geophysics (Section 5.5). The VLM read the abstracts of all 50 candidate papers and rejected both: the Phase-2 Iran-Natanz posterior is 0.058, with one surviving “weak” R&D signal whose key-phrase set explicitly includes *nuclear fuel cycle*, *uranium enrichment*, *nuclear technology*, *nuclear diplomacy*. Macro mean posterior across all four cases drops from 0.291 to 0.182, with top-1 accuracy preserved at 1.00 throughout.

Why ECE worsens under VLM activation. Under the one-bin reliability formula we use in Phase 1 ($ECE = \frac{1}{n} \sum_i |p_i - \mathbb{1}[\text{top1 correct}]|$), the heuristic’s noise-driven over-confidence on Iran was *closer* to the binary truth label of 1.0 than the VLM’s calibrated under- confidence is. The metric, as defined, rewards confident-and-correct over uncertain-and-correct, so a clean model can score worse on 1-bin ECE than a noisy one. We treat this as a flaw of the metric (and of single-bin ECE in small- n regimes) rather than of the VLM. A more discriminating Phase 2 calibration analysis will use bucketed reliability diagrams against the imagery and trade modalities once they ramp up evidence per case.

Cost and rate. The full four-case VLM run consumed roughly \$0.30 of OpenRouter credit and completed in 90 seconds wall-clock. Per-paper VLM classification took 1.5–3 seconds; this is well within the budget for the Phase 2 expansion to twenty-plus retrofit cases.

5.7 GNN-light leave-one-out evaluation

We extract a 9-dimensional feature vector per state from the APA graph (node degree, evidenced-process count, average SUGGESTS strength, path counts to Pu / LEU / U-233, best-path

strength to Pu, cell-volume breadth, and material-edge count) and train a logistic regression classifier with leave-one-out cross-validation across the four cases. A synthetic indicator-stripped negative is paired with each positive, giving a $4+4 = 8$ -sample dataset.

Table 4: GNN-light LOO results. Mean p_{positive} over the positive folds vs. over the synthetic negative folds. Both classifiers attain perfect LOO accuracy; GBM separates the distributions more sharply.

Classifier	LOO accuracy	Mean p_+ on positives	Mean p_+ on negatives
Logistic regression	1.00	0.869	0.080
Gradient boosting	1.00	0.961	0.003

Caveat. The eight-sample dataset and synthetic indicator-stripped negatives mean these LOO numbers should be read as a sanity check on the feature pipeline rather than a calibrated risk estimate. A realistic evaluation requires negatives from declared peaceful programmes with comparable indicator counts, which is a Phase 2 task once the imagery and trade collectors have ramped up evidence per case. We report GNN-light here to make the feature engineering reproducible and to provide a head-to-head reference for the eventual GCN/GraphSAGE upgrade.

5.8 Cost and reproducibility

The Phase 1 run consumed only public OpenAlex requests (~ 2 s for 50 papers) and zero VLM tokens. The end-to-end wall-clock for one retrofit case was 3.0 s (excluding OpenAlex network latency). All Phase 1 results are reproducible via `python -m scripts.natanz_retrofit --no-vm` from the released code archive; the Phase 2 VLM run will require an `ANTHROPIC_API_KEY` (estimated cost $\sim \$2$ per case with prompt caching enabled).

6 Discussion

6.1 What is recoverable from open sources

The Phase 1 evidence (Section 5.5) supports the following partial answer to the central question of this paper. Auxiliary infrastructure (E08), the R&D fingerprint (E05), and trade bundles (E04) are recoverable to a useful degree from public sources; the Iran-Natanz retrofit, even when restricted to OpenAlex literature hits before the 2002 disclosure, isolated the correct process volume (V03 Enrichment) from a 109-paper haystack down to a small E05 fingerprint. EDE (E01) is partially recoverable when imagery resolution is sufficient: this is the principal Phase 2 question once the Sentinel-2/Landsat-7 retrieval is enabled.

6.2 What is not recoverable

- **Indicator strength weights** as defined by IAEA expert judgment (we provide a Bayesian + Cooke-elicitation surrogate but note it is not equivalent).
- **Weaponization indicators (PM Vol. 11)** — intentionally out of scope for ethical reasons.
- **Classified ground truth** from inspections, environmental swipes, or member-state declarations.

6.3 Risks of misuse and mitigation

The Phase 1 false-positive on geophysical *centrifuge* papers illustrates the most likely misuse mode — naive consumers might treat threshold-only system passes as substantive findings. We

mitigate through (i) explicit per-indicator provenance recorded in **evidence** fields, (ii) advisory-committee sign-off on indicator-strength rubrics before any case publication, (iii) academic license restriction in **LICENSE** and a stated discouragement of operational use in **README**, and (iv) Vol. 11 (weaponization) exclusion enforced at the schema level (cells in V03/V07 carry an explicit **redacted_per_vol11_policy** flag where applicable). The April 2026 Planet Labs imagery restriction in Iran further underscores the political fragility of OSINT pipelines and reinforces our preference for free Sentinel-2/Landsat as the baseline.

6.4 Relation to JRC and Niemeyer-group work

The JRC “Big Table” [12, 11] populates a similar indicator/process matrix from open sources but does so in tabular form and without VLM-native multimodal extraction; OS-PM is best viewed as a VLM-native successor that re-uses the JRC schema intuition while adding a graph-native APA layer. The Niemeyer-Listner formal APA [8, 1] contributes the verification correctness framework into which OS-PM’s posteriors plug as evidence weights. We do not duplicate either programme; rather, OS-PM provides a publicly reproducible automation layer beneath both.

7 Conclusion

We have presented OS-PM, a VLM-driven, open-source, partial reconstruction of the IAEA Physical Model. On the first retrofit case (Iran-Natanz, 2002 cut-off, literature-only) the system isolated the correct enrichment volume from a noisy candidate set, but did so via a literal-token false positive that the planned VLM upgrade is designed to remove. We see OS-PM as a *methodological contribution* for academic state-level safeguards research, not a substitute for the IAEA’s own evaluation process. We release code, prompts, and 54 of 96 PM-schema YAML cells under Apache-2.0 to invite community extension; the schema is complete (88/88 in-scope element cells across 11 volumes). Phase 2 will add three further retrofit cases, imagery, and the VLM ablation.

Acknowledgments

[Funder, advisory committee, beta testers; to be filled at submission time. Anthropic API academic credits if applicable. We deliberately do not name individual IAEA personnel without explicit permission.]

Ethical statement

This research was reviewed and approved by [Institution]’s IRB and a dedicated dual-use review committee (decision [ID]). All data is publicly licensed; no classified or member-state-restricted material was used. Outputs related to weaponization (PM Vol. 11) are intentionally excluded by design.

Data and code availability

- Code: <https://github.com/pinking-jeong/Nonproliferation> (Apache-2.0).
- PM cell schema (YAML): [\[zenodo-doi\]](#).
- Retrofit datasets: [\[zenodo-doi\]](#).
- All experiments are reproducible via `make reproduce`.

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